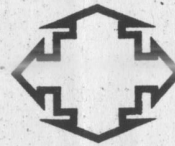


Roll No. :



BCA-5003

BCA (Semester-V) (NEP) Examination, 2025-26

COMPUTER NETWORK

Time : 2 Hours]

[Maximum Marks : 75

Note : 1. Attempt questions from **all** sections as directed.

2. The candidates are required to answer only in serial order. If there are many parts of a question, answer them in continuation.

3. "B" copy will not be provided.

Section-A

(Short Answer Type Questions)

Note : There are **nine** questions in this section. Attempt **all** questions. Each question carries **5** marks. [$9 \times 5 = 45$]

1. (a) What is Communication?

- (b) Mention the advantages of Optical Fiber Cables.
- (c) Explain the use of Bridges.
- (d) What is Piggybacking?
- (e) Write short note on CRC.
- (f) What are the Network Components? Explain.
- (g) Define Error Control.
- (h) What is a Protocol?
- (i) Discuss the services offered by the Application Layer.

Section-B

(Long Answer Type Questions)

Note : There are **four** questions in this section. Attempt **any one** question. Each question carries **15** marks.

[1×15=15]

- 2. Explain the functions of TCP/IP protocol.
- 3. Write short notes on the following :

- (a) Packet Switching
- (b) Message Switching
- (c) Circuit Switching

- 4. Describe the sliding window protocol for data link layer.
- 5. What is Framing? Explain different types of framing protocols with their format.

Section-C

(Long Answer Type Questions)

Note : There are **four** questions in this section. Attempt **any one** question. Each question carries **15** marks.

[1×15=15]

- 6. Explain the applications of computer networks.
- 7. Explain ISDN architecture and its services.
- 8. Explain the cryptography used for network security and privacy.
- 9. Describe the congestion control algorithms.

-----X-----